

Assignment 2: IP Addressing and Subnetting

[40 mks]

Subnetting Scenario

Scenario 1: 192.168.2.0/24 → 4 subnets

To get 4 subnets, borrow 2 bits ($2^2 = 4$): new mask **/26 = 255.255.255.192**. Each subnet has $2^6 = 64$ addresses → **62 usable hosts** per subnet.

Subnet	Network Address	Usable Host Range	Broadcast Address
1	192.168.2.0/26	192.168.2.1 – 192.168.2.62	192.168.2.63
2	192.168.2.64/26	192.168.2.65 – 192.168.2.126	192.168.2.127
3	192.168.2.128/26	192.168.2.129 – 192.168.2.190	192.168.2.191
4	192.168.2.192/26	192.168.2.193 – 192.168.2.254	192.168.2.255

Scenario 2: 30 usable hosts, starting at 10.0.0.0

Need $2^n - 2 \geq 30 \rightarrow n = 5$ host bits ($2^5 - 2 = 30$). Mask: **/27 = 255.255.255.224**

- **Network address:** 10.0.0.0/27
- **Usable host range:** 10.0.0.1 – 10.0.0.30
- **Broadcast address:** 10.0.0.31

Scenario 3: 172.16.0.0/20

Assuming this is a Class B network (default /16) subnetted to /20:

- Borrowed bits: $20 - 16 = 4 \rightarrow$ **16 subnets**
- Host bits: $32 - 20 = 12 \rightarrow 2^{12} - 2 =$ **4,094 usable hosts** per subnet
- Mask: 255.255.240.0 (block size of 16 in the 3rd octet)

Subnet	Network Address	Broadcast Address
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1	172.16.0.0/20	172.16.15.255
2	172.16.16.0/20	172.16.31.255
3	172.16.32.0/20	172.16.47.255

VLSM (Variable Length Subnet Mask)

Scenario: 192.168.3.0/24

Allocate largest requirement first to minimize waste.

Department	Hosts Needed	Mask	Network Address	Usable Host Range	Broadcast
Marketing	60	/26 (255.255.255.192) — 62 usable	192.168.3.0/26	192.168.3.1 – 192.168.3.62	192.168.3.63
Sales	30	/27 (255.255.255.224) — 30 usable	192.168.3.64/27	192.168.3.65 – 192.168.3.94	192.168.3.95
Engineering	10	/28 (255.255.255.240) — 14 usable	192.168.3.96/28	192.168.3.97 – 192.168.3.110	192.168.3.111

Logic: Sort requirements descending (Marketing → Sales → Engineering), then assign each department the smallest block size that satisfies $(2^n - 2) \geq \text{required hosts}$, placing each new block immediately after the previous one ends. This leaves 192.168.3.112 – 192.168.3.255 free for future growth.